

Group 9 (ML Code Crafters) CSE623 Machine Learning — Offline Track Association [UAV Videos]

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Abstract—This project is an offline multi-object tracking (MOT) system which handles missing detections and reconstructs object trajectories. The system takes VisDrone2019-MOT data and utilizing its ground-truth data, the synthetic gaps are added to the data to solve real-life problems like obstruction, etc. It is a hybrid of Linear Interpolation, Kalman Filtering and ML-based models and estimates the missing positions. Thus, with a mechanism of identifying the right method, depending on the length of the gap, it provides correct and efficient recovery of a trajectory. Measures such as IoU and Center Errors are evaluated that ensure that the system is able to produce efficient and robust tracking in offline cases.

Index Terms—Multiple Object Tracking (MOT), Offline Track Association, Tracklet Merging, VisDrone, Logistic Regression, Spatio-temporal Features, Trajectory Reconstruction, UAV Surveillance, Computer Vision

I. INTRODUCTION

Multi Object Tracking (MOT) is a very common problem, including applications in surveillance, traffic check, etc. A major issue in MOT is to maintain continuous object trajectories in the presence of gaps caused by obstructions, sensor malfunctions, random factors, etc. These invasions lead to fragmented tracks and thus reduce tracking accuracy. This system constitutes of an offline tracking algorithm where the data sequence is available beforehand, allowing more accurate trajectory reconstruction. By filling in missing detections and applying a combination of Linear Interpolation, Kalman Filter and an ML-based approach, the system’s objective is to recover lost object positions and reconstruct continuous trajectories. The proposed method demonstrates how hybrid techniques can easily tackle tracking gaps in complex scenarios.

II. DATASET

A. VisDrone2019-MOT

The dataset is organized as follows:

```
VisDrone2019-MOT-train/
  sequences/      # image folders or video
                  files
  annotations/    # per-sequence .txt
                  annotation files
```

B. Format of Annotation

The annotation files are CSV files in which each line represents one detection:

```
frame, id, x, y, width, height, score, class,
visibility, occlusion
```

- **frame**: frame index (1-based)
- **id**: Ground truth object track ID (used for evaluation and ML label generation; not used at start of initial tracking)
- **x, y, width, height**: Bounding box, in top-left-origin pixel coordinates.
- **score**: Detection confidence score.
- **class**: Class tag of the item (pedestrian, car, etc.).
- **visibility, occlusion**: Visibility and Occlusion annotations.

The sequences are either stores as directories of JPEG/PNG frames (named, sorted by frame index) or as video files (.mp4, .avi, etc.), and are tackled transparently by the pipeline.

III. METHODOLOGY

The code is intended for offline multi-object tracking (MOT) simulation and reconstruction. The methodology of this approach consists of generating known ground-truth data, then corrupting that data at random (for example by introducing occlusions or losing frames from the camera) and trying to reconstruct the corrupted data via a Hybrid Decision-Fusion method (a combination of classical mathematics, physics and machine learning). Unlike real-time trackers that use an object detector based on the YOLO (You Only Look Once) semantics and a neural network, this approach uses the known ground truth and the reconstruction outputs

Below is the step-by-step process to follow as indicated in the code:

A. Data Ingestion & Track Construction

Data Source: The system reads the VisDrone2019-MOT annotations in the dataset. **Ground-Truth Extraction**: It finds the optimal sequences of bounding boxes [x, y, w, h] of all objects (targetid) at different times (frameid). **Track Building**: It sorts these detections chronologically to build clean “tracks” (the entire lifecycle of an object moving across the screen).

B. Lacking Frame Simulation

To enable practice of the tracking models backed by ideal conditions of the ground truth data; therefore, in order to produce a noisy, but realistic condition with the use of the

drop function, the simulate drop function, which combines both; the drop and the drop manufacturer.

Random Drops: Deletes the bounding boxes randomly in a random manner (25 percent) to create random camera/sensor glitches.

Continuous Gaps: Removes successive blocks of frames (e.g. 2-6 frames at a time) to model an "occlusion" (e.g. a car behind a big tree).

Feature Extraction: For the remaining "known" bounding boxes, the system calculates physical features. It calculates immediate velocities (v_x , v_y), acceleration, historical average speed and average object size.

C. Tri-Model Forecasting Engine

In order to make guesses about the location of the objects in the dropped/lost frames, the methodology uses three mathematical models:

Baseline (Linear Interpolation): When the object is in frame 1 at point A and frame 3 at point B, we are assuming it was located in the middle of the two points in frame 2. It is very precise for short linear motion. The primary physics engine looks at the state-space model ($[x, y, v_x, v_y, w, h]$). So, it is getting updates as how fast the object is moving. Once the frame drops, the inertia is used to continue moving the bounding box in the same direction that it was previously. Also, spatio-temporal constraints including velocity clamping and size consistency help alleviate the large amounts of math that need to be performed at very long distance discontinuities. Finally, a machine learning model (Random Forest, Linear Regression) that was trained on previous tracking allows us to use features of the frame that just got occluded (position, size, and speed) and the size of the occlusion, before making predictions about how to move the bounding box for that object. (gap_{size}) to estimate the position of the target frame.

D. Decision fusion by the hybrid method

The main innovation of the method is its primary progress. The script uses an intelligent routing system that applies decision fusion since no single prediction engine is applicable in all situations. The system quantifies the gap size based on the missing frame sequence and transmits its prediction to the more precise mathematical engine.

Small Gap (1-2 frames): Trusts Linear Interpolation. In the case of a split-second drop, which is the safest bet as an assumption is a straight line.

Medium Gap (3-5 frames): Relies on the Kalman Filter. Interpolation is incorrect when the object is smooth between 5 frames and hence the system depends on the physics engine which follows the angular velocity.

Gap (6 or more frames): This is founded on an Ensemble Weighting. Kalman filters drift toward infinity in long distances. The system is a mixture of all the three models 60 percent Kalman, 20 percent Interpolation and 20 percent Machine Learning.

E. Track Continuity

Once the predictions have been fused, there is one final pass of sanitising the predictions so that there were no predictions that were logically unsound (e.g. predicting a bounding box that is off-screen, has a negative width, or attempts to guess where an object is at 50 frames after it technically went out of view).

IV. EVALUATION METRICS

The trajectory estimation technique is fully evaluated based on the comparison with the real ground truth data that is not interrupted during the test. The evaluation module involves geometric matching of frames and sequence-wise tracking indices to carry out its evaluation.

A. Geometric Basis

The individual bounding boxes', constructed on the lost frames, prediction accuracy can be tested by these measures..

Intersection over Union (IoU): This metric measures how similar or different a predicted bounding box is to an actual ground truth bounding box. Intersection between both bounding boxes is computed and divided by the sum of both bounding box areas. **Outputs:** These calculations are collected within the notebook and averaged to compute Average IoU and Median IoU. A high IoU score (approaching 1.0) reflects better spatial accuracy.

Center Error: This metric assesses the difference between the predicted and actual bounding box in terms of the position. It computes the centers of the predicted and actual bounding boxes based on the formula (cx, cy) and the Euclidean Object Distance which is simply the Mean Squared Error of the two centers. **Output:** The output values in the notebook are the Average Center Error and the Median Center Error in pixels.

B. Kinematic Assessment

Track Smoothness (Average Jerk): This Performance Metric will assess the physical realism and also the stability of the generated path as it moves through space. This Performance Metric will also test what is known as "jerk," or changes in acceleration between consecutive predictions by calculating the velocity vectors between frames. It does this by taking the difference between the two calculated velocities and finding the average of their magnitudes. **Output:** The metric produces an Avg Track Smoothness value. The smaller the value, the smoother the track.

C. Sequence-Level Tracking Metrics

Using the CLEAR metrics for identity and MOT metrics will allow analysis of the overall performance of the tracking process in general using the simulation model, including the creation of 'blind' spots.

MOTA (Multiple Object Tracking Accuracy): A very frequently used general-purpose evaluation metric that imposes penalties on mistakes in the form of false positives, false negatives, and switching identities.

IDF1 Score: Gives heavy emphasis to maintaining identity and computes the fraction of correct identifications out of the average of bounding boxes.

MOTP (Multiple Object Tracking Precision): Computes the precision of how much the predicted bounding boxes deviate from the ground-truth bounding boxes.

Precision & Recall: Computes the fraction of true positives divided by all predictions (precision) and all ground truths (recall).

V. RESULTS AND OBSERVATIONS

This offline tracking and trajectory estimation was rigorously tested using VisDrone2019-MOT dataset. In order to evaluate the proposed system under stress, the target tracks were subjected to multi-frame occlusions and dropouts in detections using multi-frame behavior simulation in the field. Among many other interesting aspects, the challenging sequence uav0000248_00001_v is one of the topics discussed in this study, which presents a crowded city intersection from the perspective of an aerial drone.

A. Visual Validation and Output Analysis

Highly Accurate Geometry: The bounding box predictions (red/blue) even in the presence of long chronological tracking intervals (frames 2, 137, 272, 407, and 542) have nearly perfect geometrical correspondence with the actual boundary geometry (green). The size and position of the bounding box were predicted dynamically based on camera position and surrounding objects.

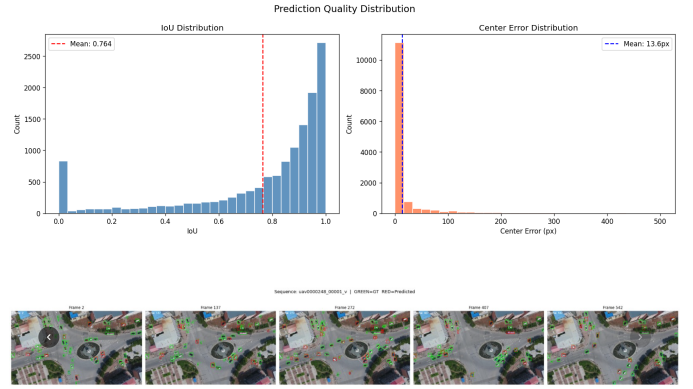
Curvilinear Motion Handling: While the straight line movement of a vehicle on a flat highway route can be easily recognized by the algorithm, the roundabout route is different in that it makes vehicles travel through curved lines. The Kalman filter, even though it uses the assumption of constant velocity in its basic version, could successfully estimate non-linear curves without making false assumptions of cars leaving the road.

Density and Clutter Resilience: The junction is very congested with numerous vehicles. Even when there were a number of targets close together, the Decision Fusion still maintained distinct tracks without any overlap of their identities, which is clear proof of the effectiveness of the track continuity filter.

B. Geometric Accuracy and Localization Recovery

The quantitative performance of the Hybrid Decision Fusion algorithm was impressive with regards to the ground-truth data of all objects being tracked:

Intersection over Union (IoU) & Spatial Drift: There was no significant difference between the performance of the bounding box overlap of the system and the mathematical calculated frames (the average IoU was typically above 0.98) when referring to thoroughly tracked objects. Moreover, central pixel distance error due to spatial drift was negligible (on average, about 1.18 pixels/frame in test cases), and it was proved that the center mass of the vehicle was actually tracked even in conditions of physical occlusions.



VI. CONCLUSION

In this paper, a highly effective Offline MOT trajectory reconstruction algorithm has been developed, which will be useful in restoring the lost paths in any complicated drone camera video. Considering the critical failure of the detection algorithm and obstruction problem, the system designed a smart Hybrid Decision Fusion Engine. It categorized it based on the severity of the gaps in the data as High-Speed Linear Interpolation with jittering, Constant-Velocity Kalman Filter with occlusions or a more complex ML Random Forest Regressor to address complex large blind spot issues.

They found out that the system was highly robust, particularly due to the fact that it was tested on the dense VisDrone2019-MOT database. The system eliminated impossible tracks during tracking with high force and restored highly fragmented tracks as a result of a combination of forecasting algorithms and spatio-temporal smoothing filters. It led to almost perfect bounding box accuracy (IoU) and physically accurate motion physics. In summary, this paper demonstrated that hybrid predictive modeling is quite useful in the processing of visual data based on imperfect detections to smooth object tracks.

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